

A behavioral genetic investigation of the Dark Triad and the Big 5

Philip A. Vernon ^{a,*}, Vanessa C. Villani ^a,
Leanne C. Vickers ^a, Julie Aitken Harris ^b

^a *Department of Psychology, Social Science Centre, The University of Western Ontario,
London, Ontario, Canada N6A 5C2*

^b *Management and Organizational Studies, Social Science Centre, The University of Western Ontario,
London, Ontario, Canada N6A 5C2*

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Abstract

This study reports the first behavioral genetic investigation of the three Dark Triad variables (narcissism, Machiavellianism, and psychopathy) and their relationships with the Big 5 personality traits. A total of 278 adult twins completed questionnaire measures of the Dark Triad and the Big 5. Consistent with some previous studies, we found significant correlations among some of the Dark Triad variables and between each of the Dark Triad variables and some of the Big 5. To the extent that these variables were correlated at the phenotypic level, these correlations were largely attributable to correlated genetic factors. At the univariate level, all traits showed the influence of genetic and non-shared environmental factors, with heritabilities ranging between .31 and .72; Machiavellianism alone also showed the influence of shared environmental factors.

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* Corresponding author.

E-mail address: Vernon@uwo.ca (P.A. Vernon).

1. Introduction

The Dark Triad is a relatively recently identified cluster of three personality traits: narcissism, Machiavellianism, and sub-clinical psychopathy (Paulhus & Williams, 2002). A narcissistic personality is characterized by an excessive love for one's self, feelings of superiority, attention seeking, and exploitativeness in relationships with others. Narcissism has primarily been studied as an abnormal personality disorder, but can also be seen as a normal aspect of personality including ideals of dominance, exploitation and feelings of entitlement (Lee & Ashton, 2005). Machiavellianism includes cold and manipulative behaviors along with insincerity and callousness. Those who score high on Machiavellianism scales tend to employ deception and manipulation for self-benefit (Jakobwitz & Egan, 2006). Psychopathy refers to a sense of high impulsivity, low remorse, and thrill-seeking (Paulhus & Williams, 2002). Like narcissism, psychopathy can be viewed from both a clinical and a non-clinical perspective. The Dark Triad of personality deals with these traits at a sub-clinical level that varies within the normal population. Previous research has demonstrated that these three traits have low to moderate correlations with one another (Jakobwitz & Egan, 2006; Lee & Ashton, 2005; Paulhus & Williams, 2002).

Correlations have also been reported between the Dark Triad and the Big 5 variables, although results have been inconsistent. Paulhus and Williams (2002), for example, reported that narcissism correlated positively with extraversion ($r = .42$) and openness ($r = .38$) and negatively with agreeableness ($r = -.36$); that Machiavellianism correlated negatively with agreeableness ($r = -.47$) and conscientiousness ($r = -.34$); and that psychopathy correlated with all Big 5 variables: extraversion ($r = .34$), agreeableness ($r = -.25$), conscientiousness ($r = -.24$), neuroticism ($r = -.34$) and openness ($r = .24$). Lee and Ashton (2005) reported similar correlations but Jakobwitz and Egan (2006) found no significant correlations between the Dark Triad and either openness or extraversion. The present study was designed to further explore relationships between the Dark Triad and the Big 5; moreover, by employing samples of twins we can estimate the extent to which any phenotypic correlations that we find between the Dark Triad and the Big 5 are attributable to correlated genetic and/or correlated environmental factors.

Numerous previous behavior genetic studies have examined the extent to which individual differences in the Big 5 are attributable to genetic and/or environmental factors. A recent review by Johnson, Vernon, and Feiler (*in press*), for example, summarized the results of 145 studies conducted between 1955 and 2007 which themselves reported twin and other kinship correlations, and heritability and environmentality estimates for the Big 5 and related traits, from a total of some 240,000 pairs of subjects. The evidence summarized by Johnson et al. (*in press*) overwhelmingly supports the conclusion that genetic and non-shared environmental factors account for the great majority of the variance, with heritabilities averaging about .45.

To our knowledge, no previous behavior genetic studies have been performed on the Dark Triad *per se* although a number of studies have reported the influence of genetic factors on psychopathy (Blonigen, Hicks, Krueger, Patrick, & Iacono, 2006; Larsson, Andershed, & Lichtenstein, 2006; Taylor, Loney, Bobadilla, Lacono, & McGue, 2003) and on narcissism (Livesley, Jang, Jackson, & Vernon, 1993). Another goal of the present study is to obtain twin correlations and to estimate heritability and environmentality coefficients for all three Dark Triad variables. We expect that we will replicate previous findings of the influence of genetic factors on psychop-

athy and narcissism and, based on previous research with other personality traits, we expect also to find a heritable component to Machiavellianism.

2. Method

2.1. Participants

Participants were 75 pairs of monozygotic (MZ) twins (58 female pairs, 17 male pairs) and 64 pairs of dizygotic (DZ) twins (50 female pairs, 14 male pairs) ranging from 17 to 92 years of age ($M = 41.42$, $SD = 17.54$). These twins were a subset of a larger sample which had taken part in a previous study (Mackie, Vernon, Martin, Sheppard, & Harris, 2006; Vernon, Mackie, Petrides, Harris, & Bacher, 2006) and who were contacted by phone or email and invited to participate in the current project. They reside across Canada and the United States. Over 95% of the twins who were contacted agreed to take part in the current study.

2.2. Materials

NEO-PI-R. This self-report questionnaire measures the factors of agreeableness, openness to experience, conscientiousness, extraversion, and neuroticism through 240 5-point items (McCrae & Costa, 1987). The test measures six sub-ordinate dimensions (or facets) of each of the Big 5 factors. It has proven to be one of the most accepted measures of the Big 5 (Jang, McCrae, Angleitner, Riemann, & Livesley, 1998).

The Narcissistic Personality Inventory (NPI). The NPI (Raskin & Hall, 1979) is used to measure sub-clinical narcissism through a forced-choice 40-item questionnaire (Paulhus & Williams, 2002). The test has demonstrated good psychometric properties (Jakobwitz & Egan, 2006).

MACH-IV. This scale (Christie & Geis, 1970) is a measure of Machiavellianism consisting of 20 5-point Likert items. Studies have determined that this test has good psychometric properties (Hansen & Hansen, 1991; Wrightsman, 1991).

The Self-Report Psychopathy Scale-III (SRP III-R12). This scale measures sub-clinical psychopathy and consists of 62 5-point items. The scale has proven to be reliable (Paulhus & Williams, 2002).

2.3. Procedure

Twins who agreed to participate in the current project were mailed the four questionnaires to be completed at their own homes. Once the questionnaires were completed, they were asked to mail them back in a stamped-addressed envelope that they were provided with. Participants were informed that by mailing back the questionnaires, they were giving their consent to participate in the present study. Upon receipt of their completed questionnaires, participants were sent \$20.00 and their names were entered into a draw to win one of ten \$100.00 prizes. Each participant was also sent a debriefing form that provided them with details of the study as well as thanks for their participation.

3. Results

The majority of twins completed all items on all the questionnaires but a few twins left one or two items blank. To account for missing items, total raw scores on the questionnaires were divided by the number of completed items to yield mean raw scores on each questionnaire. No significant differences existed between MZs and DZs on these mean raw scores and MZ and DZ variance/covariance matrices were computed from these scores and used in the following analyses using un-weighted least squares. Given the wide age range of the sample and the uneven numbers of males and females, all data were corrected for age and sex using McGue and Bouchard's (1984) regression approach prior to the following analyses.

Shown in Table 1 are the MZ and DZ twin correlations and genetic and environmental parameter estimates derived from univariate behavioral genetic model-fitting analyses. a^2 represents the proportion of the variance in each variable attributable to additive genetic effects, while c^2 and e^2 represent the proportion of the variance attributable to shared and non-shared environmental factors, respectively. Although MZ correlations are more than twice as large as DZ correlations for some variables we did not attempt to estimate dominance genetic effects (d^2) due to the lack of power arising from our relatively small samples (Martin, Eaves, Kearsley, & Davies, 1978).

As can be seen in Table 1, MZ correlations are larger than DZ correlations for all variables and the model-fitting results indicate that for all variables except Machiavellianism the majority of the variance is attributable to genetic and non-shared environmental factors, with genetic factors accounting for between 31% and 72% of the variance. Machiavellianism has a heritability of only .31 and shared environmental factors contribute moderately highly (.39) to its variance. There is a very small (and non-significant) contribution of the shared environment to psychopathy and to agreeableness; for the other variables the shared environment has no effect. The relatively small sample sizes limit the power to detect significant parameter estimates and this is reflected in their 95% confidence intervals, several of which contain zero. It may also be noted, however, that shared environmental effects are almost never detected in adult twin personality data even when sample sizes are large enough to detect these if they were present (Johnson et al., *in press*).

Shown in Tables 2 and 3 are the phenotypic correlations (rp) among the Dark Triad variables and between the Dark Triad and the Big 5 variables. Also shown in these tables are the results of

Table 1
Twin correlations and parameter estimates for the Dark Triad and Big 5 variables

Variable	<i>r</i>		Parameter estimates (95% CI)		
	MZ	DZ	a^2	c^2	e^2
Narcissism	.52	.33	.59 (.12–.71)		.41 (.29–.58)
Machiavellianism	.68	.57	.31 (.00–.76)	.39 (.00–.69)	.30 (.20–.46)
Psychopathy	.54	.44	.64 (.16–.80)	.04 (.00–.42)	.32 (.20–.51)
Openness	.73	.27	.72 (.36–.81)		.28 (.19–.43)
Conscientiousness	.58	.19	.58 (.20–.72)		.42 (.28–.63)
Extraversion	.51	.16	.50 (.06–.67)		.50 (.33–.72)
Agreeableness	.66	.36	.63 (.14–.78)	.04 (.00–.46)	.33 (.22–.50)
Neuroticism	.62	.06	.57 (.25–.71)		.43 (.29–.63)

Note. a^2 = additive genetic effects; c^2 = shared environmental effects; e^2 = non-shared environmental effects; CI = confidence interval.

Table 2

Multivariate genetic analyses of the narcissism, Machiavellianism, and sub-clinical psychopathy scales

	Narcissism	Machiavellianism
Machiavellianism	rp = .12 rg = – rc = .18 (–.05–.39) re = .14 (–.11–.38) AIC = –2.99	
Sub-clinical psychopathy	rp = .28** rg = .48 (.27–.66) rc = – re = .20 (–.07–.44) AIC = 1.89	rp = .51** rg = .66 (.48–.81) rc = – re = .27 (.02–.49) AIC = –6.33

rp = phenotypic (observed) correlation; rg = genetic correlation; rc = shared environmental correlation; re = non-shared environmental correlation. Numbers in brackets represent the 95% confidence interval values. * $p < .05$.

AIC = Akaike Information Criterion.

** $p < .01$ (two-tailed).

multivariate model-fitting analyses which include genetic correlations (rg) and shared (rc) and non-shared (re) environmental correlations between the variables and these correlations' 95% confidence intervals. Akaike Information Criterion (AIC) goodness-of-fit statistics are also reported: larger negative values indicating a better fit.

First, at the phenotypic level, narcissism correlates positively with extraversion, openness to experience, and psychopathy, and correlates negatively with agreeableness. Machiavellianism correlates positively with neuroticism and psychopathy, and correlates negatively with agreeableness and conscientiousness. Psychopathy correlates negatively with agreeableness and conscientiousness and, as mentioned, correlates positively with Machiavellianism.

The multivariate model-fitting results indicate that the phenotypic correlations reported above are primarily attributable to correlated genetic and, to a lesser extent, to correlated non-shared environmental factors. Any of these correlations whose 95% confidence interval does not include zero are significant at the .05 level. Significant genetic correlations were found between: narcissism and psychopathy, extraversion, openness to experience, and agreeableness; Machiavellianism and psychopathy, agreeableness and conscientiousness; and psychopathy and agreeableness and conscientiousness. Only two non-shared environmental correlations reach significance: those between Machiavellianism and neuroticism and between Machiavellianism and psychopathy. There are no significant shared environmental correlations between any of the variables, indicating that shared environmental factors do not contribute at all to any of the phenotypic correlations.

Multiple regression analyses were performed to estimate the amount of variance in each of the Dark Triad variables that could be accounted for by the Big 5 variables in combination. These analyses yielded adjusted R^2 s of .18 for narcissism ($F = 10.96$, $p < .01$), .25 for Machiavellianism ($F = 16.70$, $p < .01$), and .39 for psychopathy ($F = 30.92$, $p < .01$). Multivariate behavioral genetic analyses revealed that these multiple correlations are also attributable to correlated genetic and correlated non-shared environmental factors: with genetic correlations (and 95% CIs) of .48 (.26–.65), .78 (.58–.96), and .85 (.71–.98) and non-shared environmental correlations (and 95%

Table 3

Multivariate genetic analyses of the NEO personality scale scores with narcissism, Machiavellianism, and sub-clinical psychopathy

	Narcissism	Machiavellianism	Sub-clinical psychopathy
NEO scale			
Neuroticism	rp = .15 rg = .03 (–.23 to .30) re = –.18 (–.43 to .08) AIC = –3.80	rp = .23** rg = .18 (–.09 to .41) re = .32 (.07–.52) AIC = –14.12	rp = .13 rg = .03 (–.27 to .30) re = .22 (–.04 to .45) AIC = 2.07
Extraversion	rp = .36** rg = .42 (.16–.65) re = .19 (–.08 to .43) AIC = –6.90	rp = –.06 rg = –.08 (.20 to –.38) re = .04 (–.21 to .29) AIC = –12.04	rp = .05 rg = –.02 (.29 to –.34) re = .11 (–.16 to .36) AIC = –0.14
Openness to experience	rp = .30 rg = .29 (.06–.49) re = .14 (–.12 to .39) AIC = –6.89	rp = .06 rg = .05 (–.18 to .28) re = .02 (–.23 to .26) AIC = –14.52	rp = .01 rg = –.11 (.14 to –.35) re = .11 (–.15 to .36) AIC = –8.66
Agreeableness	rp = –.37** rg = –.42 (–.20 to –.42) re = –.13 (–.12 to –.14) AIC = –6.03	rp = –.49** rg = –.70 (–.52 to –.85) re = –.21 (.04 to –.43) AIC = –18.63	rp = –.59** rg = –.78 (–.61 to –.91) re = –.33 (–.09 to –.54) AIC = 2.10
Conscientiousness	rp = –.03 rg = –.07 (–.33 to .19) re = –.03 (–.24 to .29) AIC = –6.22	rp = –.32** rg = –.43 (–.18 to –.67) re = –.14 (.12 to –.38) AIC = –19.19	rp = –.37** rg = –.50 (–.24 to –.72) re = –.16 (.10 to –.40) AIC = –4.44

rp = phenotypic (observed) correlation; rg = genetic correlation; re = non-shared environmental correlation. Numbers in parentheses represent the 95% confidence interval values. * $p < .05$.

AIC = Akaike Information Criterion. *Note:* all relationships were best fit by an AE model.

** $p < .01$ (two-tailed).

CI) of .29 (.06–.50), .16 (–.09 to .40), and .30 (.06–.52) for narcissism, Machiavellianism, and psychopathy, respectively.

4. Discussion

This study had three main goals: first, to examine correlations among the 3 Dark Triad variables and between these variables and the Big 5; second, to examine the extent to which individual differences in the Dark Triad are attributable to genetic and/or environmental factors; and thirdly to examine the extent to which observed phenotypic correlations between the Dark Triad and the Big 5 are themselves attributable to correlated genetic and/or correlated environmental factors.

Consistent with previous studies, we found significant positive correlations between narcissism and psychopathy and between Machiavellianism and psychopathy. These correlations support the notion that the Dark Triad consists of a cluster of at least somewhat correlated personality traits. On the other hand, we did not find a significant correlation between narcissism and Machiavel-

lianism, which is inconsistent with Paulhus and Williams (2002) and with Jakobwitz and Egan (2006), which both reported moderate correlations between these variables, but is consistent with Lee and Ashton's (2005) report of a non-significant correlation. Our conclusion is that although the three Dark Triad traits conceptually appear to measure somewhat overlapping dimensions of personality, empirical support for this overlap is not always forthcoming.

Also consistent with some previous studies, we found significant correlations between each of the Dark Triad and several of the Big 5 variables. Like Paulhus and Williams (2002), we found narcissism correlating positively with extraversion and openness and negatively with agreeableness. Our study and Paulhus and Williams (2002) also found negative correlations between Machiavellianism and agreeableness and conscientiousness; we also found that Machiavellianism correlated positively with neuroticism. Our results for psychopathy are only partially consistent with those of Paulhus and Williams (2002): we only found significant (negative) correlations between psychopathy and agreeableness and conscientiousness. The main conclusion to be drawn here is that although there are some correlations between the Dark Triad and the Big 5, the fact that these are neither large nor consistent indicates that the two do represent overlapping but distinct clusters of personality. At the multivariate level, the Big 5 variables in combination accounted for between 18% and 39% of the variance in the 3 Dark Triad variables, again indicating only a moderate amount of overlap.

MZ and DZ twin correlations and univariate behavioral genetic model-fitting analyses reveal that, as has been found in previous studies, psychopathy and narcissism both have a moderate-to-large heritable component. Machiavellianism was only somewhat heritable and, unlike all the other variables in this study, showed a more pronounced influence from the shared environment. The model-fitting results for the Big 5 confirm that individual differences in these traits are entirely attributable to genetic and non-shared environmental factors. The fact that the results for the Big 5, narcissism and psychopathy are in line with previous behavioral genetic studies suggests that the results for Machiavellianism are not simply a fluke due to an unusual sample. Rather, it appears that Machiavellian-like behaviors are to some extent learned in addition to being transmitted genetically. It should also be noted that although Livesley et al. (1993) reported dominance genetic effects contributing to narcissism (as measured by the DAPP-BQ), whereas our study only modeled additive genetic effects, their sample size was also relatively small and likely lacking the power required to properly detect dominance (Martin et al., 1978). Our studies also employed different measures.

From the multivariate analyses it can be concluded that observed correlations among the Dark Triad and between the Dark Triad and the Big 5 are largely attributable to the influence of the same genes. Several of the genetic correlations are quite large – for example those between psychopathy and agreeableness (–.78) and between psychopathy and Machiavellianism (.66) – indicating considerable overlap between the genes influencing these traits. The influence of correlated non-shared environmental factors was less pronounced – only two of these correlations reaching significance – and shared environmental factors did not contribute to the correlations between any of the traits. This latter finding and the finding of significant genetic correlations is consistent with much previous personality research.

In conclusion, this study reports the first behavioral genetic investigation of the Dark Triad as a cluster and is also the first to investigate the extent to which genetic and environmental factors contribute to relationships among and between the Dark Triad and the Big 5. With the exception

of Machiavellianism, our results are consistent with those of previous behavioral genetic studies of other personality traits which have demonstrated the primary importance of genetic and non-shared environmental factors in contributing both to individual differences on the traits and to correlations between them.

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